Sangyu Han

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Research Summary

My research focuses on explainable AI and mechanistic interpretability for deep neural networks. I study how internal concepts are formed, represented, and propagated across layers, with the aim of making model behavior more transparent and understandable. More recently, I have been particularly interested in practical interpretability for model debugging, using interpretability methods to analyze failure cases, detect spurious behavior, and improve the reliability of AI systems.

Education

- **Seoul National University**, Seoul, Republic of Korea — *Aug. 2022 – Present*
  Integrated M.S./Ph.D. Program in Artificial Intelligence
- **Korea University**, Seoul, Republic of Korea — *Mar. 2016 – Aug. 2022*
  B.S. in Health and Environmental Science
  Interdisciplinary Major in Artificial Intelligence

Publications

Conference and Workshop Publications

1. **Bi-ICE: An Inner Interpretable Framework for Image Classification via Bi-directional Interactions between Concept and Input Embeddings**
   Jinyung Hong*, Yearim Kim*, Keun Hee Park, **Sangyu Han**, Nojun Kwak, Theodore P. Pavlic.
   WACV, 2026
2. **VDPP: Video Depth Post-Processing for Speed and Scalability**
   Daewon Yoon*, Injun Baek*, **Sangyu Han**, Yearim Kim, Nojun Kwak†.
   CVPR 2026 ECV Workshop, 2026
3. **Causal Interpretation of Sparse Autoencoder Features in Vision**
   Sangyu Han, Yearim Kim, Nojun Kwak.
   ICCV 2025 eXCV Workshop, 2025
4. **Respect the model: Fine-grained and Robust Explanation with Sharing Ratio Decomposition**
   Sangyu Han*, Yearim Kim*, Nojun Kwak.
   ICLR, 2024
5. **Feature extraction on Convolutional Neural Network Layers with Channel-Wise Neuron Vector**
   Sangyu Han, Gilsang Yoo.
   International Conference on Innovation Convergence Technology, 2020

International Journals

6. **From Local to Global to Mechanistic: An iERF-Centered Unified Framework for Interpreting Vision Models**
   Yearim Kim*, **Sangyu Han***, Nojun Kwak†.
   IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2026
7. **Beyond Activation Patches: ERF-Grounded Autolabeling of Sparse Autoencoder Features in Vision Transformers**
   Sangyu Han*, Yearim Kim, Nojun Kwak†.
   NeurIPS 2026, under review

* equal contribution   † corresponding author

Teaching Experience

- **Samsung AI Expert Program, Teaching Assistant** *Sep. 2024 & 2025*
  Conducted teaching assistance for "Explainable and Responsible AI" in the Samsung AI Expert Program, including the 7th cohort in 2025.
- **Korea Data Agency, Teaching Assistant** *Jun. 2021 – Sep. 2021*
  Assisted with the "Curriculum on Intelligent Information System Development" project.

Projects

- **Predicting academic performance of students by revealing socio-latent features, Korea University** *Dec. 2021 – Jun. 2022*
  Developed a predictive model for academic performance using 6 years of poll data.
- **Sleep stage classification from non-EEG PSG data using long short-term memory neural networks, Korea University Medical Center** *Dec. 2021 – Jun. 2022*
  Predicted sleep stages using non-EEG polysomnography data from MAXIM smart band.
- **Development of a stereo image-based environmental recognition algorithm for lunar exploration mobility, Hyundai Motor** *Jul. 2023 – Jul. 2024*
  Generated bird's-eye-view maps of lunar surfaces with stereo-based 3D object detection.

Awards

- **Dean's List, Korea University** *2nd Semester, 2020*
  Honors for students with a cumulative GPA of at least 4.50/4.50.

Skills

- **Languages:** Korean (Native), English (Advanced)
- **Technical Skills:** Python (PyTorch, NumPy), C++